

AL-2001

Artificial Intelligence

Lab # 9

Objectives:

- NumPy Arrays

Note: Carefully read the following instructions (*Each instruction contains a weightage*)

1. First think about statement problems and then write your logic on Copy / Notebook.
2. Write **Your Name** and **Roll No** on your Paper/Sheet's first page.
3. **Do not copy from any source otherwise you will be penalized with negative marks.**
4. Complete your lab **within given Time Slot**.
5. Paste all your codes along with screenshots in a word file and renamed with your roll number.
6. Keep all your source files in your computer for verification. Do not overwrite a single source file for all programs.

Problems:

1. Create a NumPy array with values from 0 to 9.
2. Create a 3x3 matrix with random values.
3. Perform element-wise addition, subtraction, multiplication, and division on two NumPy arrays.
4. Calculate the mean, median, and standard deviation of a NumPy array.
5. Slice a 1D array to get elements at even indices.
6. Use boolean indexing to extract elements from an array that satisfy a certain condition.
7. Reshape a 1D array into a 2D array.
8. Stack two NumPy arrays horizontally and vertically.
9. Perform element-wise matrix multiplication.
10. Find the maximum and minimum values in a NumPy array with their indexes.
11. Calculate the sum of elements along rows and columns in a 2D array.



12. You have a CSV file containing historical stock prices with the following columns: "Date," "Open," "High," "Low," "Close," and "Volume." You need to perform various operations on this data using NumPy.
1. Load the CSV data into a NumPy array.
 2. Calculate the daily returns for the "Close" prices.
 3. Find the date on which the stock had the highest trading volume.
 4. Determine the average closing price for Mondays and Fridays.
 5. Identify the top three days with the largest intraday price fluctuations (use the difference between the "High" and "Low" prices).
13. You are working with scientific data that's organized in a NumPy array. The data represents temperature measurements taken at different locations over time.
1. Calculate the mean, standard deviation, and variance of the temperature data.
 2. Identify the location with the highest and lowest average temperature.
 3. Determine the year with the highest temperature variance.

You need to do your exercise within the given time.